CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. (IT) Examination

December 2012

IT403 Data Compression

Date: 04.12.2012, Tuesday | **Time: 10:00 a.m. To 01:00 p.m.** | **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Choose the appropriate word/s to fill in the blanks and justify your choice. **[07]**

1. Data Compression = ___________+___________. (Building , Modeling, functions, Encoding, Decoding, Coding, Equations)
2. Extended Huffman Coding is preferred when probability distribution is _______________. (Uniform , Skewed, Gaussian).
3. Huffman Coding is ___________ coding technique.(Top-Down, Bottom-up).
4. Shannon-Fano Coding is _________ coding technique.(Top-Down, Bottom-up).
5. Source entropy H becomes maximum for an alphabet containing N distinct symbols, when probabilities are _________for all symbols.(uniform, distinct, skewed)
6. _________ and ________ algorithms generate a codeword for a pattern of symbol instead of a symbol.(Shannon-Fano , Extended Huffman ,Arithmetic Coding ,Non-Binary Huffman ,Minimum variance Huffman)
7. Modeling is a process of building model of ___________.(Data, Redundancy, Information)

Q - 2 Attempt following questions. **[14]**

(a) Write an algorithm for checking Uniquely decodable property of a code. Check following codes for Uniquely Decodable property. **4**

1.{1,01,000,001} 2. {1,01, 10, 0100,1000,10011 }

(b) Explain performance measurement parameters of Data Compression algorithms. **5**

(c) Define terms: Self Information, Entropy, Compression Ratio, Redundancy removed Average Length. Also provide the relationship among them. **5**

OR

Q - 2 Attempt following questions. **[14]**

(a) Differentiate: 1. Lossy vs. Lossless technique **4**

2. Static vs. Adaptive technique

(b) A Source α={a,b} with probabilities as {0.91 ,0.09}. Find the extended Huffman code for order-0 and order-1. Find Entropy H, Average Length L_{AVG}, Compression Ratio for **5** each case. Comment on the results obtained.

(c) If the probabilities of all symbols are in order of 2^{-N}(N is an integer), then Huffman coding performs best compression.(True/False). Justify your answer with suitable example. 5

Q - 3 Attempt following questions. **[14]**

(a) A Source emits 8 symbols with probabilities 1/4, 1/4, 1/4, 1/16, 1/16, 1/16, 1/32 and 1/32 respectively. Find the entropy of the source. Obtain compact 4-ary Huffman code and find the average length of the code. Determine the redundancy of the code. 4

(b) In arithmetic coding range get decreases as the symbols is encoded. What are the possible solutions to overcome this problem? What are the purpose of using E_1, E_2 and E_3 mappings? 5

(c) It is possible that more than one code can exist for an alphabet containing n symbols. Justify your answer with suitable example. 5

OR

Q - 3 Attempt following questions. **[14]**

(a) Mention the advantages and disadvantages of Data Compression. 2

(b) Generate Arithmetic coding tag for string "CSPIT". 5

(c) Assume our alphabet consists of the 26 lowercase letters of the English alphabet. Encode the string "**ATTRIBUTE**" using Adaptive Huffman coding. Show the encoded message. 7

SECTION – II

Q - 4 Match the correct pairs and justify. **[07]**

A	B
1. Quantization Table	A. Pattern encoded as an index
2. LZW	B. Spatial to Frequency domain Conversion
3. Sampling	C. sampling amplitude axis
4. Discrete Cosine Transform	D. Image Quality Control
5. Nyquist Rate	E. Pattern encoded as a Triplet
6. Quantization	F. Sampling time axis
7. LZ77	G. Sampling Frequency

Q - 5 Attempt following questions. **[14]**

(a) How the token size is calculated in sliding window dictionary based algorithm? How does the window size affect the performance of dictionary based algorithms? Explain with suitable examples. 4

(b) Consider an alphabet α={A, B, O, W, _, eof} and corresponding CODE={1,2,3,4,5,6}. Compress following string using LZ78 dictionary based algorithm. 5

String: WABBA_WABBA_WABBA_WABBA_WOO_WOO_WOO<eof>

(c) Given the following initial dictionary and the received sequence below, build an LZW dictionary and decode the transmitted sequence. Initial dictionary as follows: 5

Index	Entry
1	A
2	C
3	G
4	T

Received Sequence: 1, 2, 1, 1, 3, 1, 4, 3, 2, 6, 4, 11, 4, 13, 18.

OR

Q - 5 Attempt following questions. **[14]**

(a) Compare and Contrast : LZ77 vs. LZ78 4

(b) Consider an alphabet α={A, B, C, D, E, eof} and corresponding CODE ={1,2,3,4,5,6} respectively. By taking window size =8 and search buffer size =4, decode following LZ77 tokens: <0,0,5>, <0,0,3>, <1,1,4>, <3,3,2>, <3,1,1>, <2,4,1>, <4,4,3>, <2,1,2>, <1,1,3>, <2,2,1>, <2,1,6>, 5

(c) Perform compression of given string using LZW algorithm and Initial dictionary given below. **String:** OHLALALALALAOHLEOHOOHLALALALALEOH 5

Index	Entry
1	A
2	E
3	H
4	L
5	O

Q - 6 Attempt following questions. **[14]**

(a) What is DCT? What do you mean by High frequency and low frequency component in an image? Why DCT in used in JPEG Image Compression? 3

(b) Draw and Explain Block Diagram of JPEG Image Compression standard. 4

(c) Explain the process of Analog to Digital conversion of analog audio signal. Discuss various parameter which affects the size of digital audio. 7

OR

Q - 6 Attempt following questions. **[14]**

(a) An ADC Device is operating at 44.1KHz and 14-Bit resolution. Find the size of audio signal for 1 minute duration. Which is the maximum frequency in audio signal that is accurately sampled and reconstructed using this device? 4

(b) What is Companding ? What is Codec? Explain Companding Codec. 5

(c) What is Quantization? What is Quantization Table? How it is important in JPEG compression? 5

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